Lecture 13

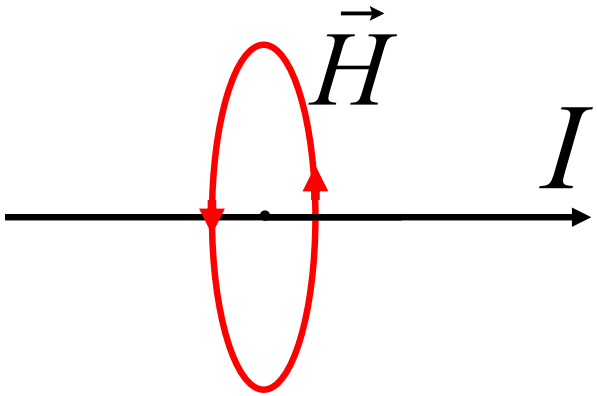
Ampere's Law

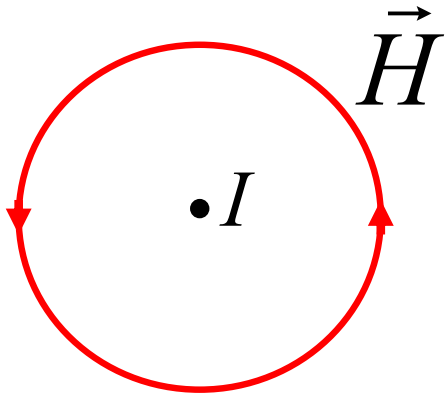
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"Static" Ampere's Law

Using the Biot-Savart Law, we solved for the **B** field around a straight wire and expressed it as **H**:





$$\vec{B} = \frac{\mu_0 I}{2\pi r}\hat{a}_\phi$$

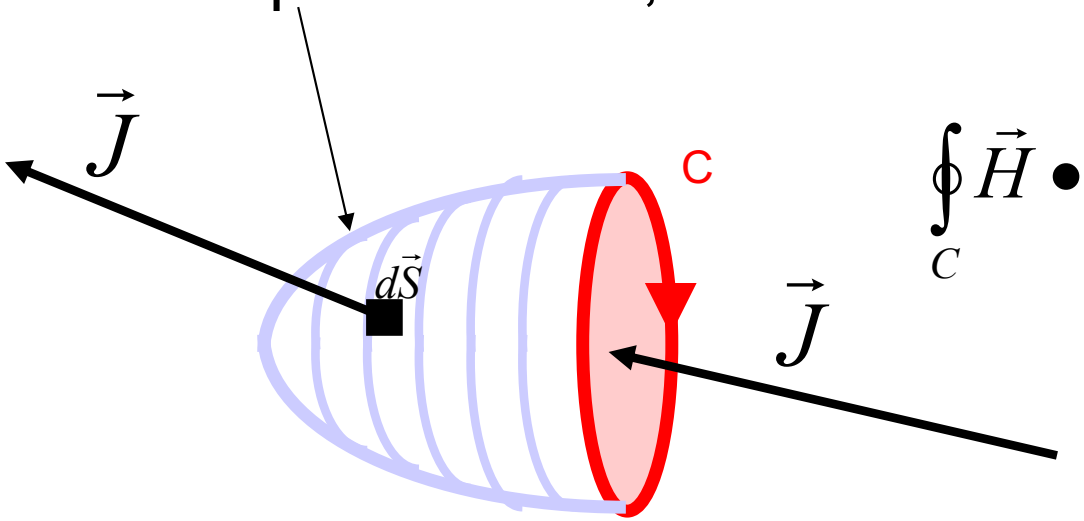
Magnetic flux density
(Wb/m^2)

$$\vec{H} = \frac{I}{2\pi r}\hat{a}_\phi$$

Magnetic field
(A/m)

Static Ampere's Law

Open Surface, where the Path forms a "Mouth"

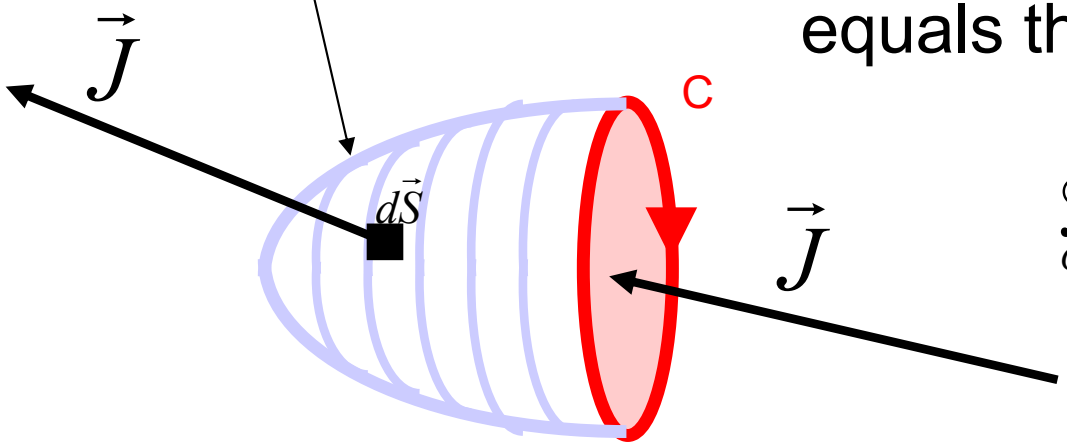


$$\oint_C \vec{H} \bullet d\vec{l} = I_{enclosed} = \iint_S \vec{J} \bullet d\vec{S}$$

In general, all the current entering the "mouth" will end up passing out of any surface that is formed around the mouth

Ampere's Law Physical Meaning

$$\oint_C \vec{H} \bullet d\vec{l} = I_{enclosed}$$



Taking the integral of **H** around a closed path equals the enclosed current.

$$\oint_C \vec{H} \bullet d\vec{l} = Magneto\,Motive\,Force\,(MMF)$$

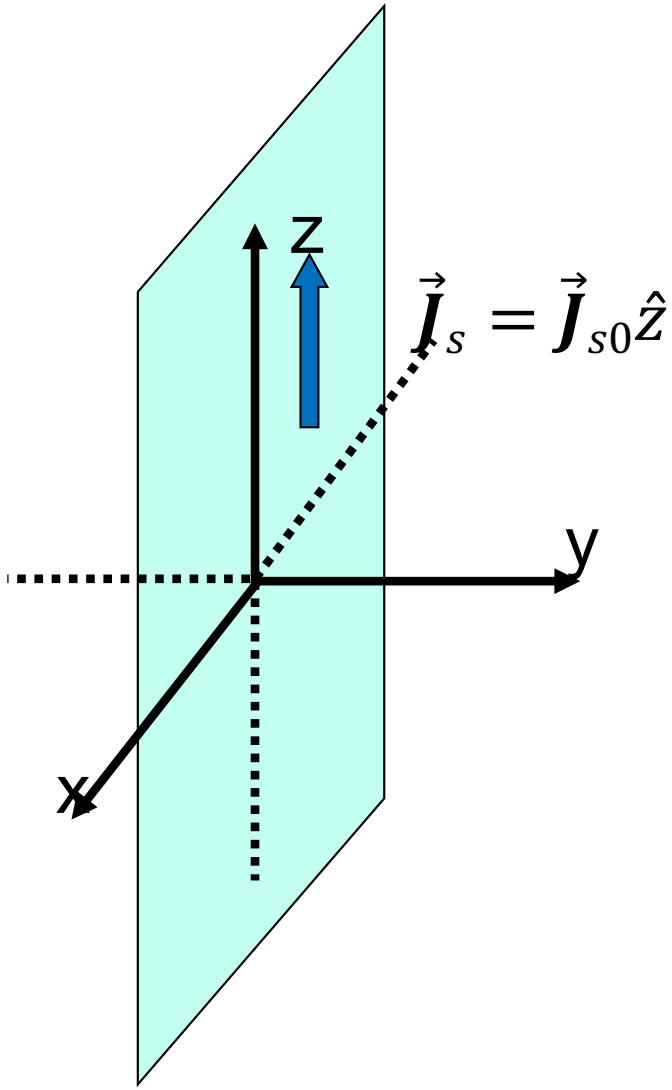
similar to:

$$\oint_C \vec{E} \bullet d\vec{l} = Electro\,Motive\,Force\,(EMF)$$

Example: Infinite Plane Sheet of Current

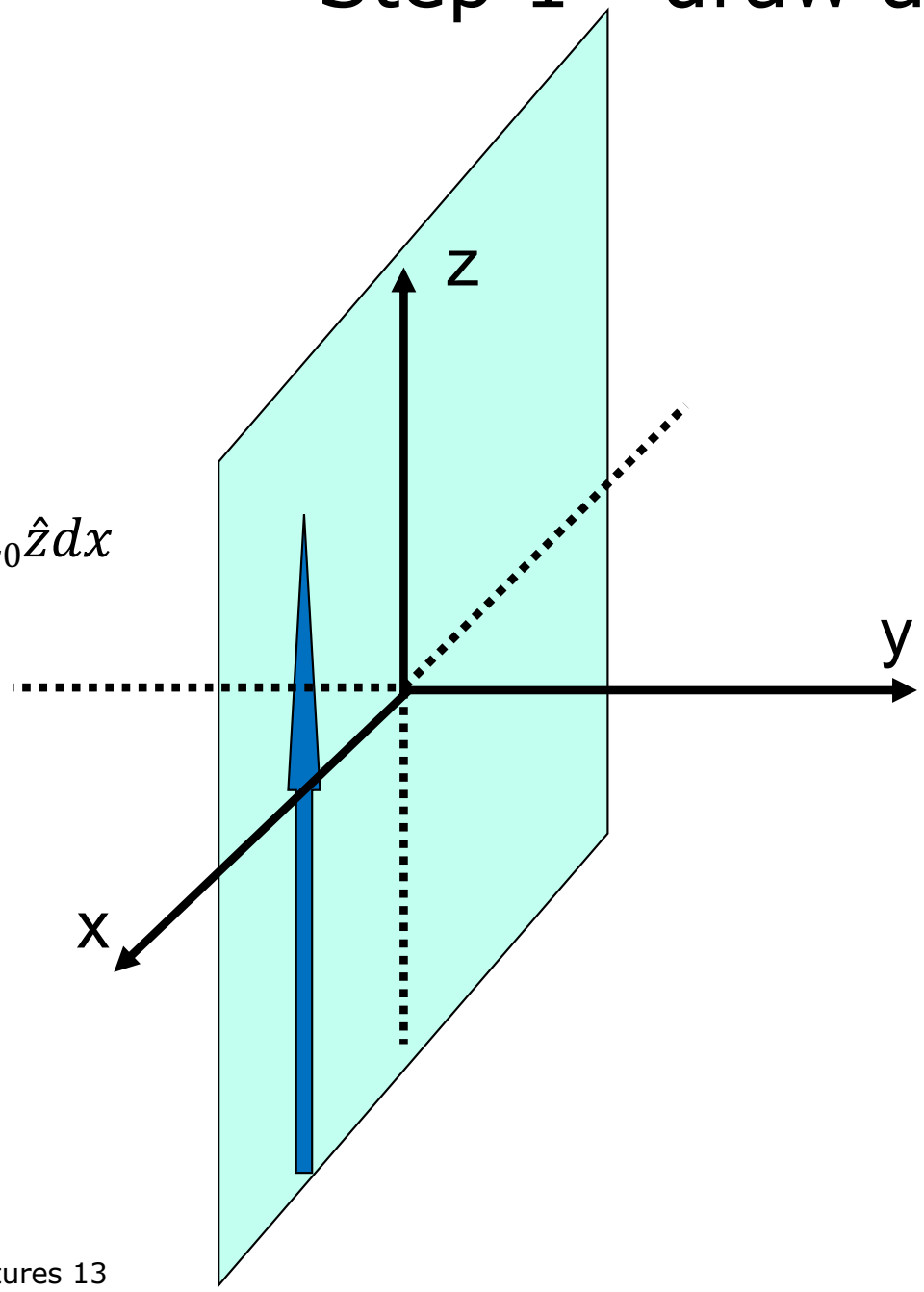
Surface current density $\vec{J}_s = \vec{J}_{s0}\hat{z}$ (A/m) flowing on $y = 0$ plane.

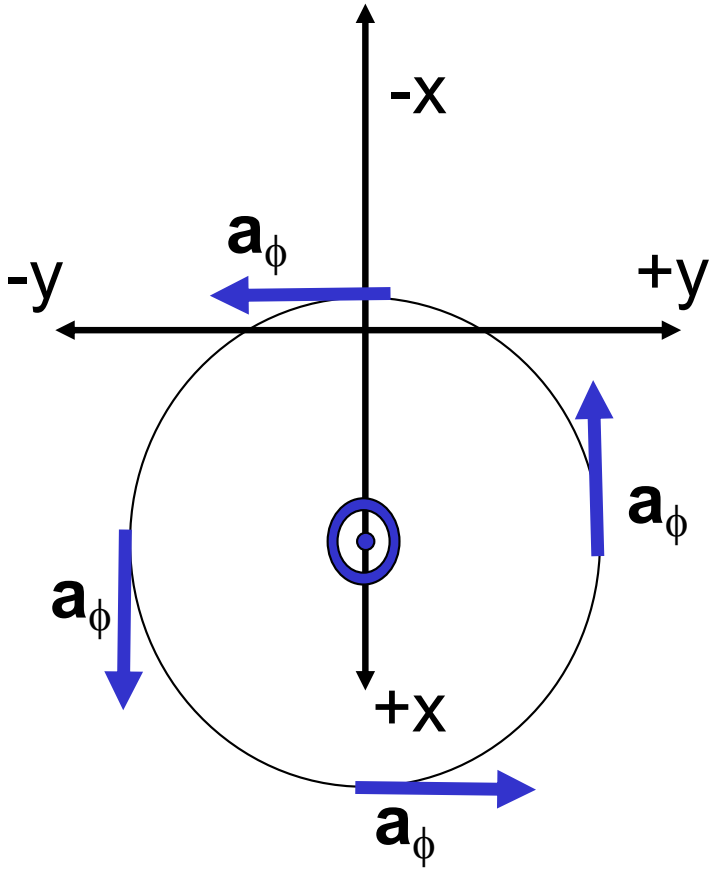
Solve for magnetic field $\vec{B}$ and $\vec{H}$



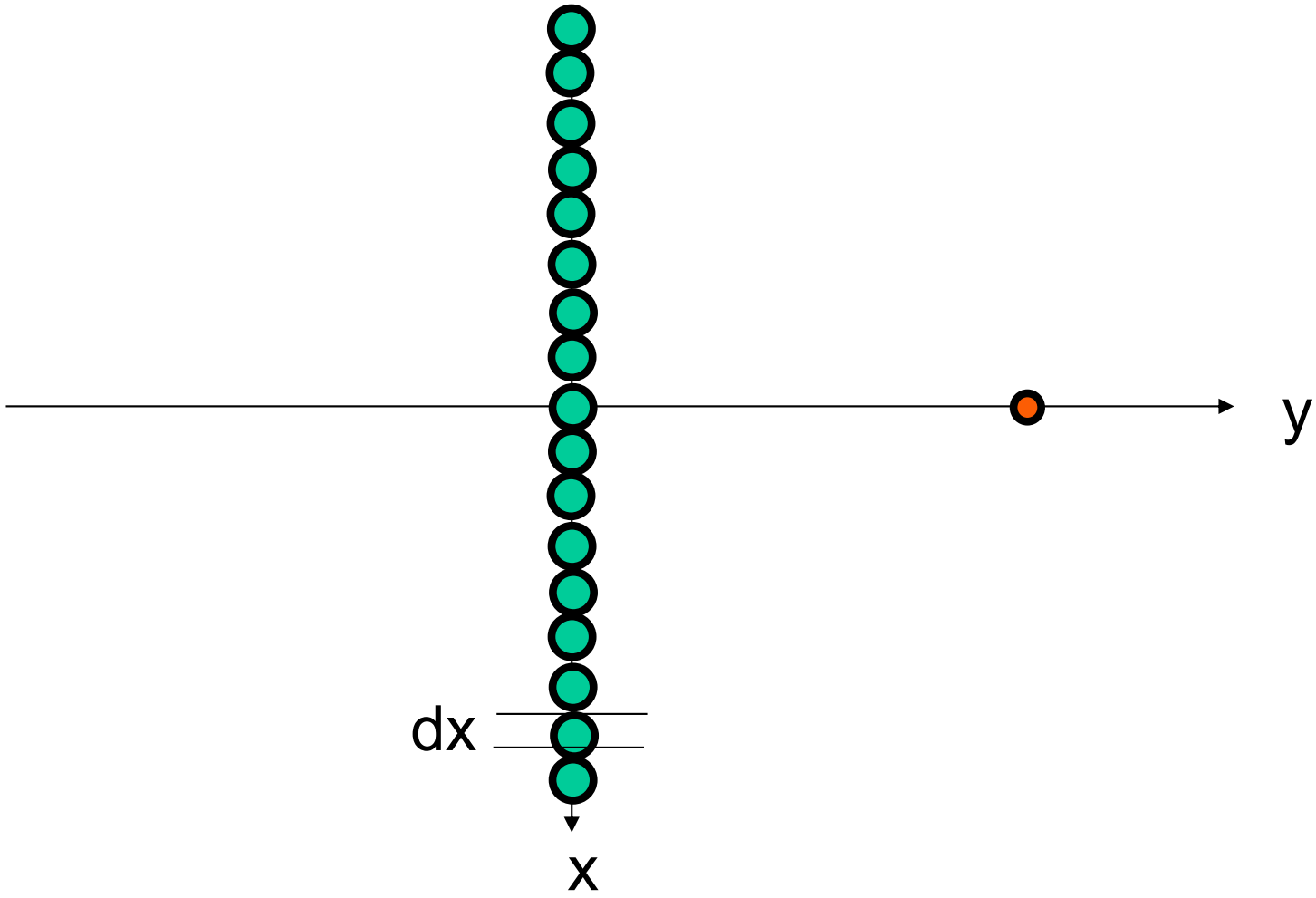
Step 1 - draw a picture

$$\boldsymbol{I}_s = \vec{\boldsymbol{J}}_{s0}\hat{z}dx$$

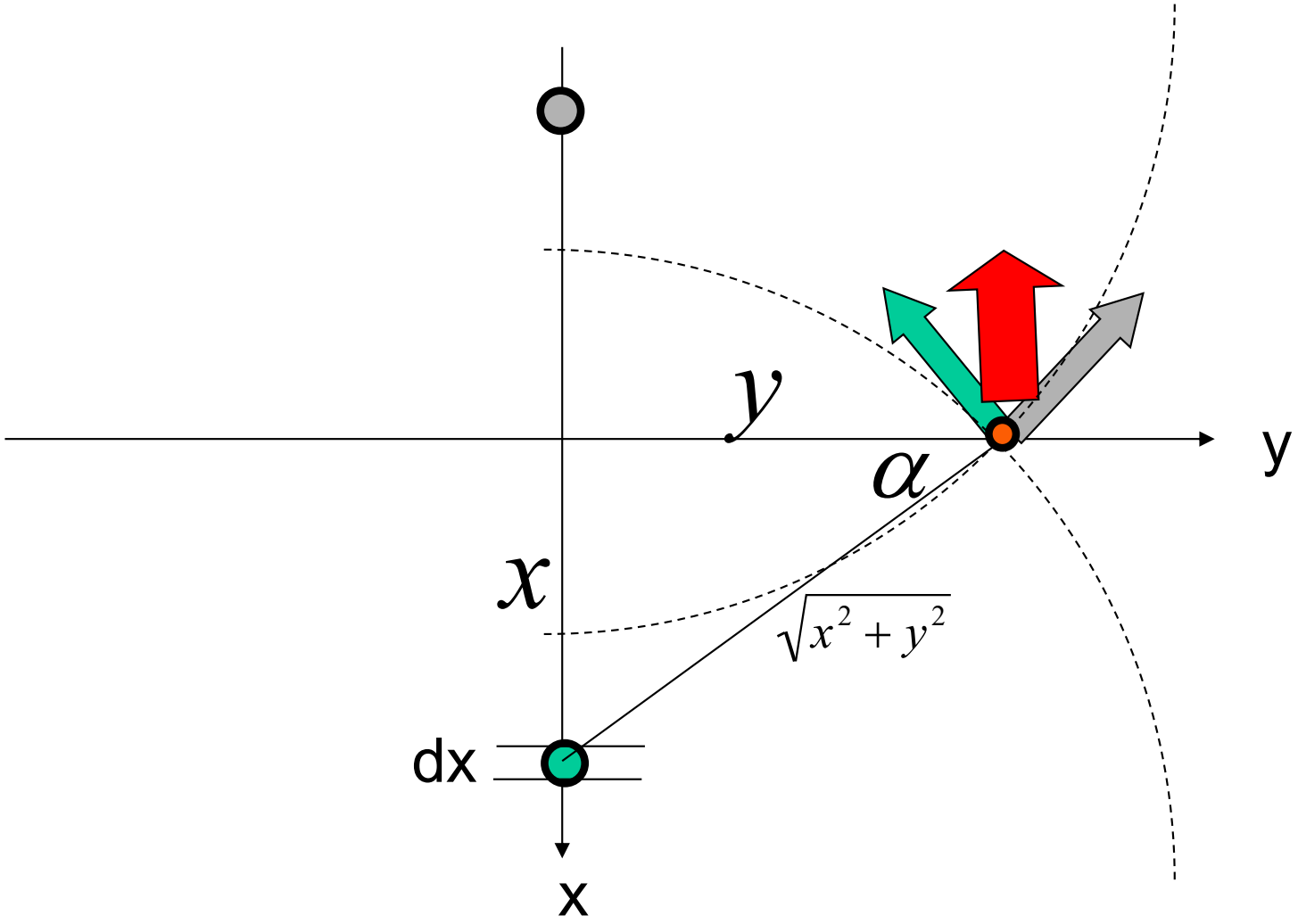




Step 2 - divide into segments

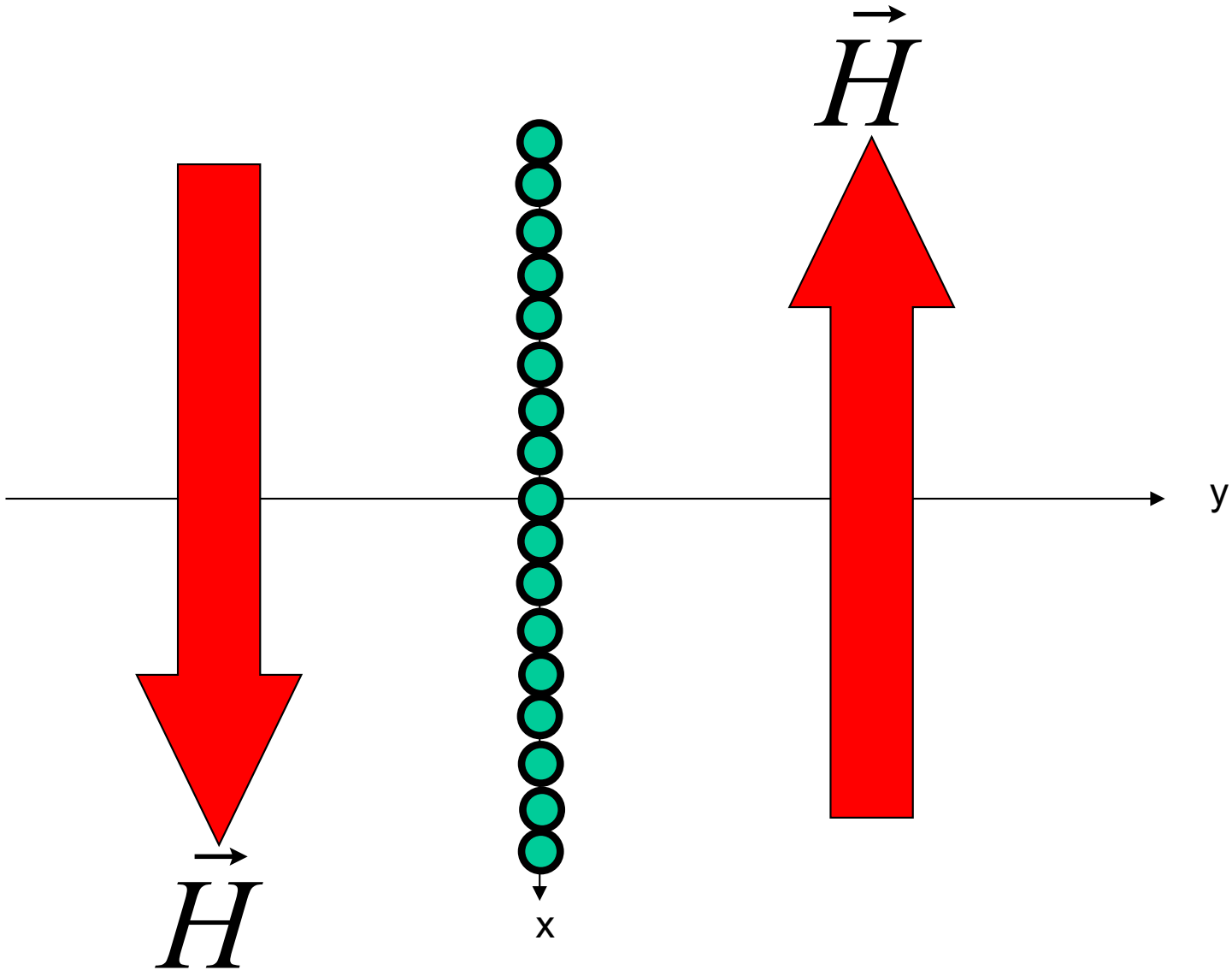


Step 3 - find d**H** for one small segment

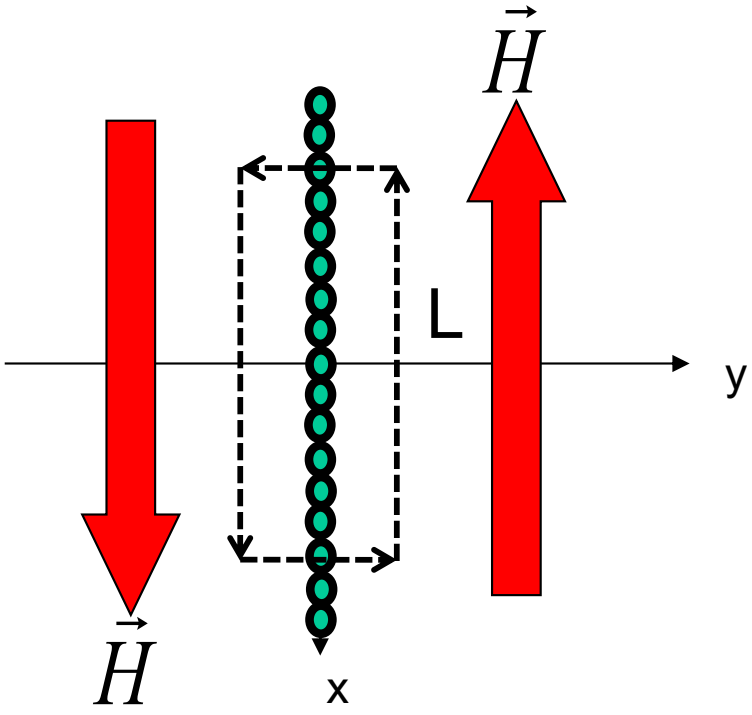


At every point (x,y), only the x-component remains, why?
For y>0, it is in $-\mathbf{a}_x$ direction

Step 4 – use symmetry to eliminate components that are zero



Aha! Now that we know H is along a_x, we can apply Ampere's Law to a simple path



$$\vec{J}_s = \vec{J}_{s0}\hat{z}$$

$$\oint_C \vec{H} \bullet d\vec{l} = I$$

$$\Rightarrow \quad |H| \cdot L + |H| \cdot L = J_{s0} L$$

$$|H| = \frac{1}{2} J_{s0}$$

$$\vec{\boldsymbol{H}} = -\frac{1}{2} J_{s0} \hat{x} \mathrm{sgn}(y)$$

$$\vec{\boldsymbol{B}} = -\frac{\mu_0 J_{s0}}{2} \hat{x} \mathrm{sgn}(y)$$

General Formula for **H** due to infinite current sheet

The direction of current flow can be in ANY direction
The current sheet might not be on a coordinate plane

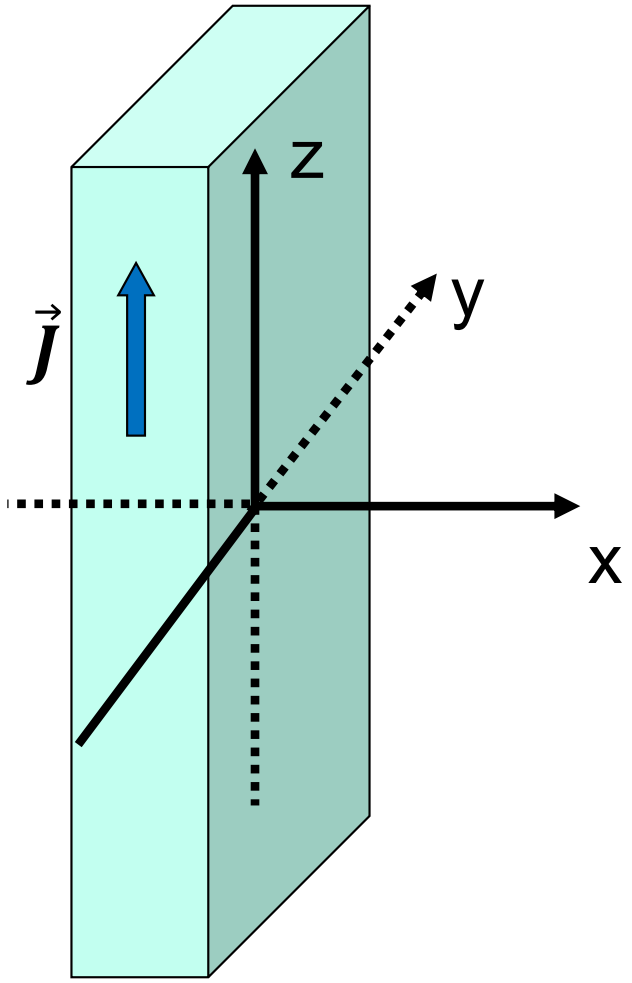
$$\vec{H} = \frac{1}{2}\vec{J}_s \times \hat{a}_n$$

Unit vector normal to the surface

Example: Infinite Slab of Volumetric Current

Volumetric current density $\vec{J}$ (A/m^2) flowing over $-\frac{W}{2} < x < \frac{W}{2}$.

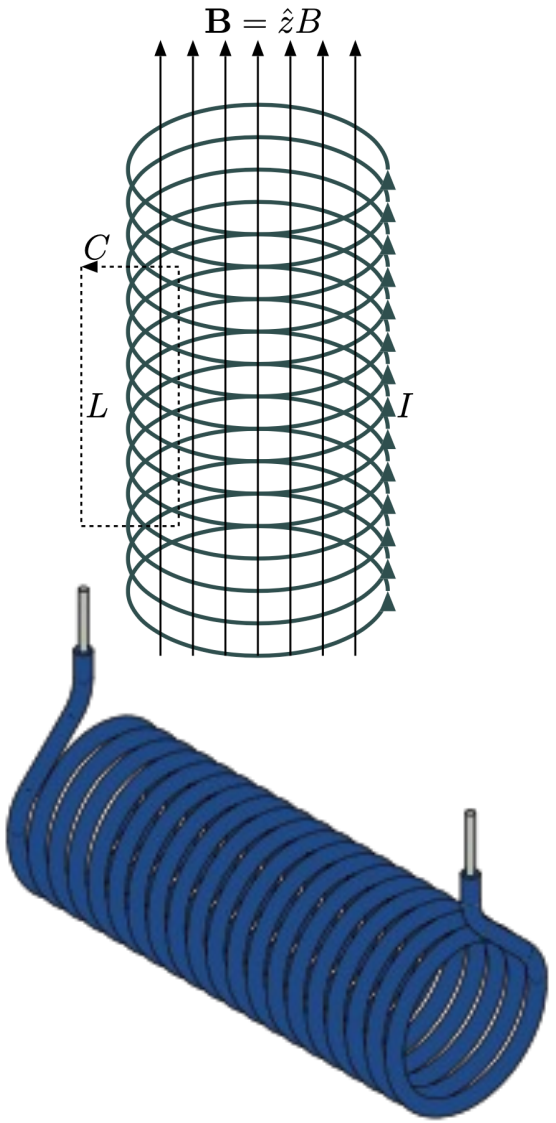
Solve for magnetic field $\vec{B}$ and $\vec{H}$



Example: Infinite solenoid

Infinite solenoid with n loops per unit length [1/m] carrying I amps per loop. The magnetic field outside of the solenoid is 0.

Solve for magnetic field $\vec{B}$ and $\vec{H}$

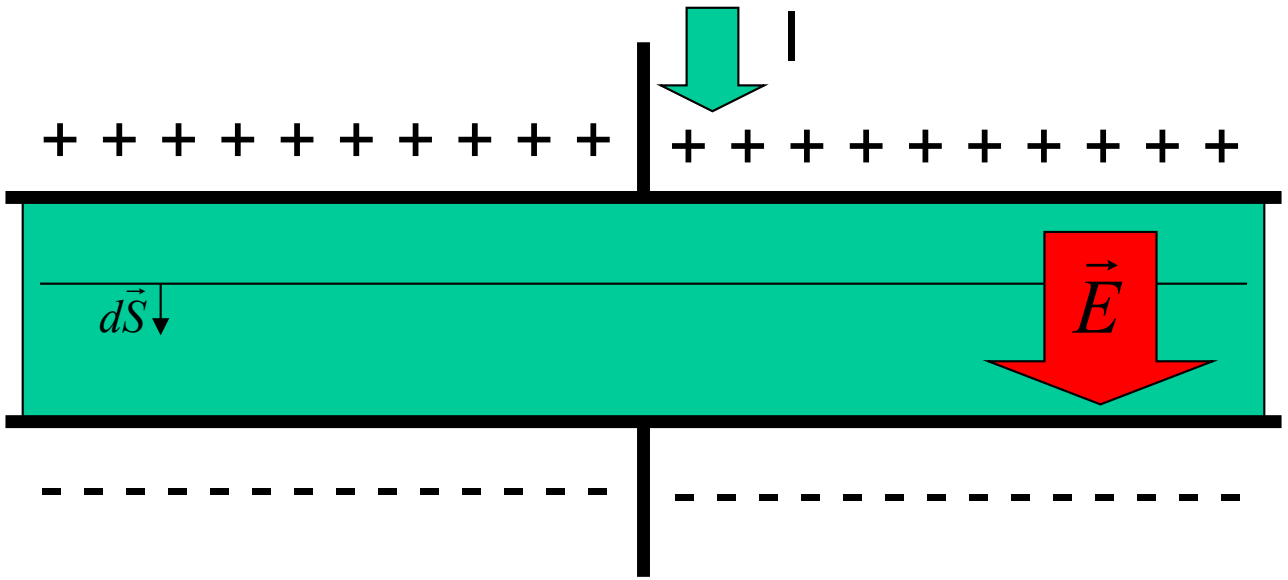


Problem with Static Ampere's Law

MMF $\oint_C \vec{H} \bullet d\vec{l} = I_{enclosed}$

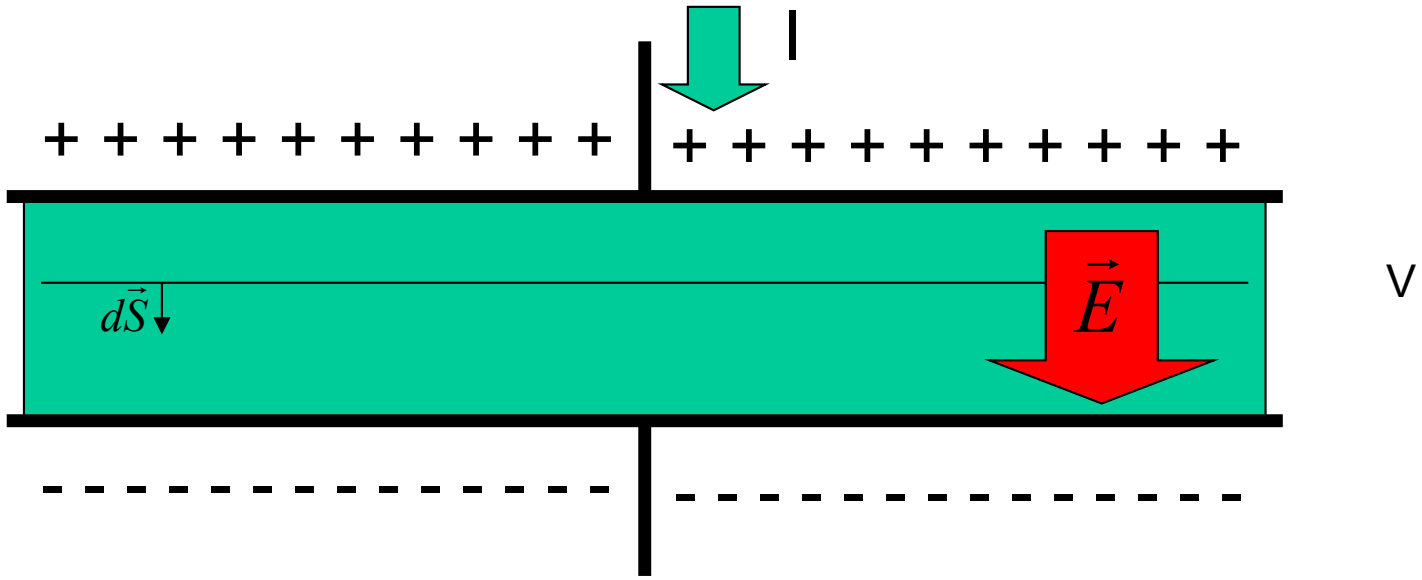
Consider a simple capacitor

- No DC current goes through
- But, AC voltage results in current flow



How does AC current get through a capacitor if it is not conducted through by a wire?

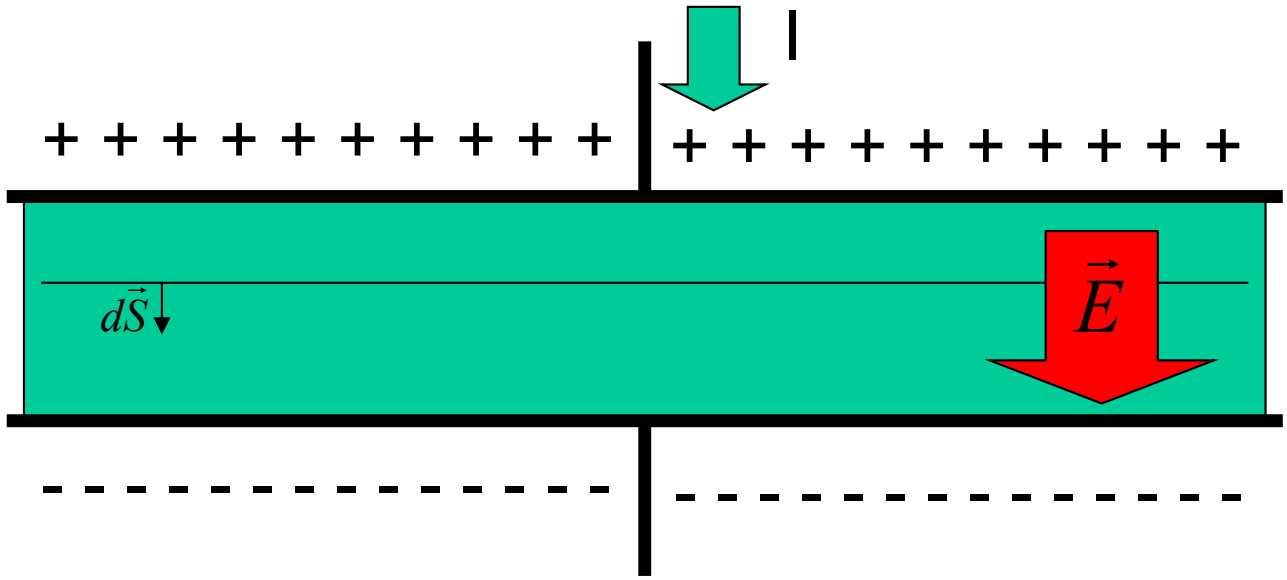
Problem with Static Ampere's Law



There is a second "way" to get current to flow

Somehow, a TIME-VARYING E field results in current flow

Problem with Static Ampere's Law



Flux of E field lines crossing a surface, S

$$\psi_E = \iint_S \varepsilon_0 \vec{E} \bullet d\vec{S}$$

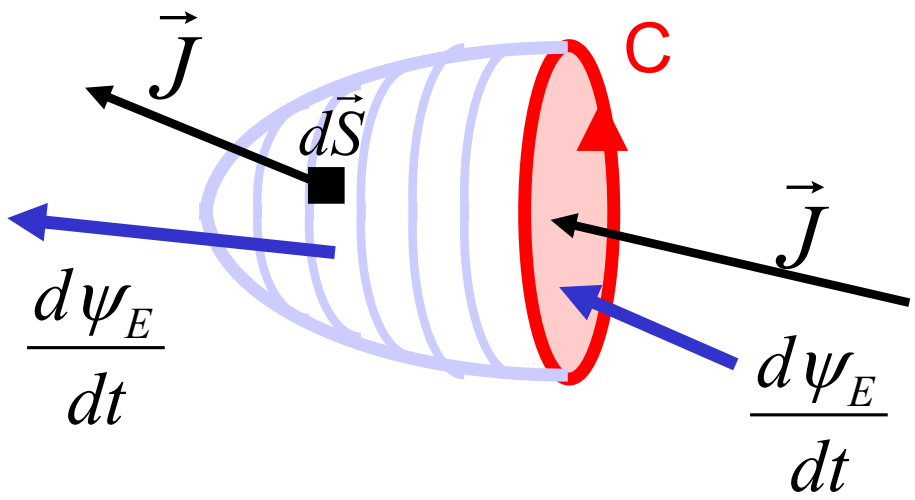
Recall the Definition:
Electric Flux ψ_E Units: (C)

James Clerk Maxwell's Genius Breakthrough

There are TWO sources of MMF:

1. Flow of charges due to current
2. Time-varying electric field

Called (by Maxwell)
"Displacement Current"



$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$

MMF (Amps) = "Regular" Current (Amps) + Displacement Current (Amps)

"New" Definition: Displacement Flux Density Vector

$\vec{E}$ "Electric Field Intensity Vector" V/m

$\vec{D} = \varepsilon_0 \vec{E}$ "Displacement Flux Density Vector"

Units: $\vec{D} = \dfrac{\text{Charge}}{\text{Area}} = \dfrac{C}{m^2}$

$$\psi_E = \iint_S \varepsilon_0 \vec{E} \bullet d\vec{S} = \iint_S \vec{D} \bullet d\vec{S}$$ Coul

$$\frac{d\psi_E}{dt} = \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$ Coul/sec (Amps)

Displacement current from a time varying E-field

- Displacement current

$$I_d = \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$$

- Find the displacement current crossing an area $A=0.1m^2$ in the xy plane from the –z to +z side for:

$$\mathbf{E}=E_0\ t\ e^{-t^2}\mathbf{a}_z$$

From Discussion 2.8 (p 106) of old book

Ampere's Law (Non static)

The Third Maxwell Equation

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$

MMF (Amps) = "Regular" Current (Amps) + Displacement Current (Amps)

$$\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$$

After using Stokes' theorem to convert to differential form

Lecture 13 Summary

- Ampere's Circuit Law (Static)
- Ampere's Circuit Law (Time varying)
- Examples
  - infinite line of current
  - infinite surface current
  - infinite slab of volume current
  - infinite solenoid
- Next class
  - Faraday's Law